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Detecting dark matter substructure in galaxies

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Abstract

The mass distribution of dark matter subhalos within galaxy-scale halos carry important information about the nature of dark matter itself. Two parameters—the substructure mass fraction f_{sub} and the power-law slope β of the subhalo mass function—are of particular interest, as their values differ between cold and warm dark matter scenarios. Extracting these parameters from strong gravitational lensing images is difficult because subhalo perturbations are subtle and the likelihood function for the image-generation process is intractable. We address both challenges by applying simulation-based inference (SBI), pairing a convolutional neural network (CNN) embedding with likelihood-free posterior estimation via neural ratio estimation. Using the SBI on simulated single-band images consistent with the Euclid space telescope, our pipeline with a lightweight embedding network (SimpleNet) achieves a mean absolute error of 0.005 ± 0.0001 , outperforming direct CNN regression by more than an order of magnitude. We then systematically assess the impact of observational nuisance parameters—stochastic lens redshift and source position, point spread function convolution, shot noise at four exposure times, three pixel resolutions, and varying lens ellipticity—on posterior quality. Longer exposure times yield the largest improvement, whereas realistic ellipticity distributions pose the greatest challenge. Our analysis offers practical guidance for deploying machine-learning-based substructure inference on data from upcoming wide-field lensing surveys.

1 Introduction

Cosmological N-body simulations based on the Λ cold dark matter (Λ CDM) paradigm predict that galaxy-scale dark matter halos host a rich population of smaller subhalos, remnants of the hierarchical assembly process through which large structures grow from successive mergers of smaller ones (Diemand et al., 2005). The predicted abundance of these subhalos—particularly at low masses ($\lesssim 10^9 M_\odot$)—depends sensitively on the microphysical properties of dark matter. Warm dark matter candidates, for instance, suppress the formation of low-mass subhalos relative to cold dark matter (Diaz Rivero and Dvorkin, 2020). Measuring the subhalo population therefore opens a window onto fundamental particle physics, but the task is observationally demanding because dark matter neither emits nor absorbs light and can only be traced through its gravitational influence (Bertone et al., 2005).

Galaxy-galaxy strong gravitational lensing provides one of the few avenues for probing this low-mass substructure. Light from a distant background galaxy, deflected by the gravitational potential of a massive foreground lens, forms arcs or Einstein rings whose detailed morphology is sensitive to mass clumps within or near the lens (Walsh et al., 1979; Varma et al., 2020). This sensitivity complements laboratory-based dark

matter searches (Undagoitia and Rauch, 2015; McCabe, 2010) and collider experiments (Sirunyan et al., 2017; ATLAS Collaboration, 2019), because lensing responds to the total projected mass irrespective of the dark matter particle type (Pearson et al., 2019; Jacobs et al., 2019). The subhalo population along a given line of sight can be statistically summarised by two quantities: the mass fraction f_{sub} , defined as the share of the host halo mass locked in subhalos, and the power-law slope β of the subhalo mass function, which sets the relative proportion of low-mass to high-mass subhalos (Brehmer et al., 2019).

The next generation of wide-field surveys will dramatically increase the number of known strong lensing systems. The Euclid space telescope, launched by the European Space Agency, is capable of observing gravitational lensing images at redshifts up to $z = 2$ and will cover $15,000 \text{ deg}^2$ of the extragalactic sky (Cropper et al., 2014). The Vera C. Rubin Observatory’s Legacy Survey of Space and Time (LSST) will provide multi-band coverage over more than $18,000 \text{ deg}^2$ (Ivezić et al., 2019; Pearson et al., 2019). These surveys are expected to yield tens of thousands of galaxy-galaxy strong lensing systems, creating both an opportunity and a computational challenge for substructure analysis.

Existing methods for detecting dark matter substructure in strong lenses rely on forward-modelling techniques such as gravitational imaging, in which the lens and source parameters are iteratively optimised and deviations from a smooth mass model are attributed to subhalos (Vegetti et al., 2014). While this approach has yielded individual subhalo detections, these are limited to relatively massive subhalos ($\gtrsim 10^8 M_\odot$) and have not yet constrained the slope β of the subhalo mass function at lower masses. Moreover, perturbations in the lensed image may also arise from line-of-sight halos that are not physically associated with the lens galaxy, complicating the attribution of observed perturbations to lens-plane substructure. Traditional forward-modelling approaches are also computationally expensive and scale poorly to the large datasets expected from forthcoming surveys.

Machine learning methods, particularly convolutional neural networks (CNNs), have emerged as a promising alternative for extracting physical parameters from lensing images (Hezaveh et al., 2017; Pearson et al., 2019). However, directly regressing substructure parameters from images using a CNN alone is challenging, because subhalo perturbations affect only a small number of pixels in the lensed image and are easily confounded with noise or macro-model variations.

To address this limitation, we employ simulation-based inference (SBI), a likelihood-free Bayesian framework that bypasses the need for an explicit likelihood function by learning the posterior distribution directly from simulated data (Greenberg et al., 2019; Brehmer et al., 2019; Cranmer et al., 2020; Durkan et al., 2020; Csilléry et al., 2010). Our SBI pipeline uses a two-component neural network architecture: a CNN that

extracts image features and maps them into a low-dimensional embedding via network embedding (NE) (Wang et al., 2016), followed by a ResNet-like classifier network (He et al., 2016) that estimates the posterior density from the embedded representation. Posterior samples are then obtained via Markov Chain Monte Carlo (MCMC). For the CNN component, we compare two architectures: EfficientNet (Tan and Le, 2019), a widely used architecture that balances depth, width, and resolution through compound scaling, and a lightweight network (SimpleNet) with fewer convolutional layers. A detailed description of both architectures is provided in Section 3.

Because the subhalo population parameters $\{f_{\text{sub}}, \beta\}$ are shared across all lensing images drawn from the same cosmological model, the inference is performed at the population level. Under the assumption that lensing images are independent and identically distributed (i.i.d.) given the population parameters, the joint posterior over N images factorises as $p(\theta \mid x_1, \dots, x_N) \propto p(\theta) \prod_{i=1}^N r(x_i \mid \theta)$, where $r(x_i \mid \theta)$ is the likelihood-to-evidence ratio estimated by the neural network classifier for each image. This amortised approach allows the trained network to be applied to any number of test images without retraining, and enables the posterior to sharpen as more images are incorporated.

In this work, we aim to (i) compare the performance of CNN-only regression against SBI-based posterior estimation for constraining f_{sub} and β from simulated Euclid-like lensing images; (ii) investigate the effect of the embedding network architecture on SBI performance; and (iii) systematically evaluate the impact of observational nuisance parameters—including randomness in lens redshift and source position, PSF convolution and shot noise at varying exposure times, pixel resolution, and lens ellipticity—on the accuracy of the inferred posteriors. The remainder of this paper is organized as follows. Section 2 provides the theoretical background on gravitational lensing, dark matter substructure, and Bayesian inference. Section 3 describes the simulation of strong lensing image datasets and the neural network architectures employed. Section 4 presents the results of the CNN-only method, the SBI approach with different embedding networks, and the nuisance parameter experiments. Section 5 discusses the implications of our findings, and Section 6 presents our conclusions.

2 Background

This section provides the theoretical context for our work. We introduce the principles of strong gravitational lensing, the role of dark matter substructure within the Λ CDM framework, existing methods for detecting substructure, and the foundations of Bayesian inference that underpin simulation-based inference.

2.1 Strong Gravitational Lensing

Gravitational lensing occurs when a massive foreground object deflects light from a more distant background source, as predicted by general relativity. In the strong lensing regime, the deflection is sufficient to produce multiple images, arcs, or complete Einstein rings of the background source (Walsh et al., 1979). The geometry of a gravitational lens system is described by the lens equation,

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta}), \quad (1)$$

where $\boldsymbol{\beta}$ is the true position of the source on the sky, $\boldsymbol{\theta}$ is the observed image position, and $\boldsymbol{\alpha}(\boldsymbol{\theta})$ is the deflection angle, which depends on the projected mass distribution of the lens. The characteristic angular scale of a strong lens is the Einstein radius θ_E , which defines the radius within which multiple images are formed. For a singular isothermal sphere (SIS) mass profile, the Einstein radius depends on the velocity dispersion σ_v of the lens galaxy and the angular diameter distances between observer, lens, and source.

The morphology and brightness distribution of the lensed images encode detailed information about the lens mass distribution. In particular, small-scale perturbations to the smooth mass model—caused by dark matter subhalos within or along the line of sight to the lens galaxy—produce localised distortions in the lensed arcs or Einstein ring. These distortions are typically subtle, affecting only a small number of pixels in the image, but they carry valuable information about the abundance and mass spectrum of the subhalo population (Kneib and Natarajan, 2011; Vegetti et al., 2014).

2.2 Dark Matter Substructure and the Λ CDM Model

In the Λ CDM picture, galaxies assemble through repeated mergers of smaller systems, a process that leaves behind gravitationally bound remnants—subhalos—orbiting within the host potential (Diemand et al., 2005). Cosmological N-body simulations show that these subhalos follow a mass spectrum extending from $\sim 10^{10} M_\odot$ down to the free-streaming cutoff set by the dark matter particle mass, which for standard CDM lies many orders of magnitude below current observational limits (Navarro, 1996).

The low-mass end of this spectrum is where different dark matter models diverge most strongly. Cold dark matter produces a steeply rising mass function, yielding vast numbers of low-mass subhalos. Warm dark matter, by contrast, erases structure below a characteristic scale determined by the particle’s thermal velocity, leading to a flattened or truncated mass function at low masses (Diaz Rivero and Dvorkin, 2020). An observational measurement of the subhalo mass function slope and normalisation can therefore discriminate between these scenarios and place constraints on the dark matter particle mass.

The subhalo mass function is commonly parameterised as a power law in mass,

$$\frac{dn}{d\log m} \propto m^\beta, \quad (2)$$

where $\beta < 0$ is the slope. Two key parameters characterise the population: the substructure mass fraction f_{sub} , defined as the fraction of the total halo mass contained in subhalos, and the slope β . A steeper slope (more negative β) implies a higher proportion of low-mass subhalos relative to high-mass ones. Constraining these parameters observationally is one of the primary goals of substructure studies in strong gravitational lensing (Brehmer et al., 2019).

2.3 Traditional Methods for Substructure Detection

Several approaches have been developed to detect and characterise dark matter substructure in strong gravitational lenses. The most established is the *gravitational imaging* technique, in which the lens mass model is iteratively refined and residual perturbations in the lensed images are attributed to substructure (Vegetti et al., 2014). This method has successfully detected individual massive subhalos ($\gtrsim 10^8 M_\odot$) in a handful of lens systems. However, it is limited in several respects: it is computationally expensive, requiring careful modelling of both the smooth lens mass distribution and the source light profile; it is most sensitive to relatively massive subhalos that produce detectable perturbations above the noise level; and it has not yet been able to constrain the slope β of the subhalo mass function.

An alternative approach exploits *flux ratio anomalies* in multiply-imaged quasars. In smooth lens models, the flux ratios between multiple images are predicted by the macro-model parameters; deviations from these predictions can indicate the presence of substructure (Keeton, 2001). While this method is sensitive to subhalos at different mass scales, it requires lensed quasar systems and is subject to degeneracies with the smooth lens model and line-of-sight structures.

Both traditional methods share a fundamental limitation: they rely on forward modelling with explicit likelihood functions, which becomes computationally prohibitive for large datasets and high-dimensional parameter spaces. This motivates the development of simulation-based inference approaches that can bypass the need for explicit likelihoods.

2.4 Bayesian Inference and Likelihood-Free Methods

The goal of statistical inference in this context is to estimate the posterior probability distribution $p(\theta \mid x)$ of the subhalo population parameters $\theta = \{f_{\text{sub}}, \beta\}$ given an

observed lensing image x . By Bayes’ theorem,

$$p(\theta | x) = \frac{p(x | \theta) p(\theta)}{p(x)}, \quad (3)$$

where $p(x | \theta)$ is the likelihood of observing image x given parameters θ , $p(\theta)$ is the prior distribution encoding our initial beliefs about the parameters, and $p(x) = \int p(x | \theta) p(\theta) d\theta$ is the evidence, a normalisation constant.

In many astrophysical problems, including substructure inference from lensing images, the likelihood $p(x | \theta)$ is analytically intractable. The image generation process involves numerous stochastic steps—drawing subhalo masses and positions from the mass function, ray-tracing through the lens, adding noise—making it impossible to write down a closed-form expression for the likelihood. However, it is straightforward to *simulate* images from the forward model for any given set of parameters.

Simulation-based inference (SBI) methods exploit this ability to simulate. Rather than computing the likelihood directly, SBI uses a neural network trained on simulated data to learn an approximation to the posterior distribution (or, equivalently, the likelihood-to-evidence ratio). This approach has gained significant traction across the physical sciences because it enables rigorous Bayesian inference even when the likelihood is intractable (Cranmer et al., 2020). Among SBI methods, Sequential Neural Ratio Estimation (SNRE) trains a classifier to distinguish between jointly sampled parameter–image pairs and marginally sampled pairs, thereby learning the likelihood-to-evidence ratio $r(\theta, x) = p(x | \theta)/p(x)$ (Hermans et al., 2020; Durkan et al., 2020). Once this ratio is learned, posterior samples can be obtained using Markov Chain Monte Carlo (MCMC) sampling with the learned ratio as the unnormalised posterior.

MCMC is a class of algorithms for drawing samples from a probability distribution that may be difficult to sample from directly. The key idea is to construct a Markov chain whose stationary distribution is the target posterior. At each step, a new parameter value is proposed and accepted or rejected based on the ratio of posterior densities (or, in our case, the learned likelihood-to-evidence ratios) at the current and proposed points. After a sufficient number of steps (the “burn-in” period), the chain converges to the target distribution, and subsequent samples approximate draws from the posterior. Common MCMC algorithms include the Metropolis–Hastings algorithm and Hamiltonian Monte Carlo; we use the implementation provided in the SBI (Greenberg et al., 2019), which employs a slice sampler for posterior exploration.

3 Methodology

This section describes our simulation pipeline and neural network architectures. Section 3.1 details the generation of simulated strong lensing images. Section 3.2 describes

the direct regression approach using a CNN. Section 3.3 explains how SBI is applied in combination with neural networks for posterior estimation.

3.1 Data Generation for Strong Gravitational Lensing Images

Simulated images in this paper were generated using the Lenstronomy (Birrer and Amara, 2018). The galaxy properties of simulated images are consistent with those of images recorded by Euclid, and the light profile from the background source galaxy is modelled using the Sérsic profile, where the Sérsic index is selected from a normal distribution with a mean of 4 and standard deviation of 1 (Pearson et al., 2019). The source location is selected from the pixel centre position of the image with a normal distribution having a mean of zero and standard deviation of 0.2. Early-type galaxies dominate strong gravitational lens systems, and the gravitational lens mass profile is therefore described usually by a singular isothermal ellipsoid (SIE). As a simplified version of the SIE model, the mass distributions of a singular isothermal sphere (SIS) also can be applied to the ellipsoid (Kneib and Natarajan, 2011). In this paper, we first use SIS to test our neural network model and we then test the different degrees of ellipsoid of SIE by using the best performance network. The Einstein radius is the characteristic angular scale of a strong lens system and determines the size of the region within which multiple images or arcs are formed. For the SIS profile, the Einstein radius depends on the velocity dispersion of the lens galaxy and the relative distances between observer, lens, and source (Brehmer et al., 2019):

$$\theta_E = 4\pi\sigma_v^2 \frac{D_{LS}(z_{lens}, z_s)}{D_S(z_s)} \quad (4)$$

where D_{LS} is the angular diameter distance between lens and source, D_S is the angular diameter distance to the source, and σ_v is the velocity dispersion in the lens system selected from a normal distribution having a mean of 250 km s^{-1} and standard deviation of 44 km s^{-1} (Pearson et al., 2019). z_{lens} is the lens redshift selected from log-normal distribution base 10 and scatter 0.25 dex and z_s is the source redshift fixed at 1.5. To observe the number of subhalos of interest and ensure all lensing image have subhalos on or outside the Einstein ring in simulation, following the suggestion of Brehmer et al. (2019), we set $\theta_r = 2 \cdot \theta_E + 1$, where θ_r is expected number of subhalos region we interested.

For the SIE profile, the lens mass distribution is no longer circularly symmetric but instead has an elliptical shape, which is more representative of real galaxies. We use the same definition of the Einstein radius as for the SIS profile and parameterise the ellipticity using two components that encode both the elongation and orientation of the mass distribution (Pearson et al., 2019):

$$e_1 = \frac{1 - q^2}{1 + q^2} \cos 2\varnothing, \quad e_2 = \frac{1 - q^2}{1 + q^2} \sin 2\varnothing \quad (5)$$

where the axis ratio of the mass profile q is selected from a normal distribution having a mean of 0.78 and standard deviation of 0.12 and $\varnothing$ is the orientation of the mass profile selected from 0 to 2π . We test different degrees of ellipticity by changing the standard deviation and orientation range. The way we change the ellipticity is by multiplying coefficients σ_m by the standard deviation of the Sérsic index, the axis ratio of the mass profile q , and the maximum range of orientation $\varnothing$. The coefficients σ_m are [0, 0.1, 0.25, 0.5, 1]. For example, if $\sigma_m = 0.1$, the Sérsic index is selected from $N(4, 0.1)$, q is selected from $N(0.78, 0.012)$, and $\varnothing$ is selected from 0 to 0.2π .

The distribution of dark matter can be described by the Navarro–Frenk–White (NFW) model based on an N-body simulation (Navarro, 1996), the Ruffini–Argüelles–Rueda model based on elementary particles (Argüelles et al., 2018), and self-interacting dark matter based on energy and momentum exchange (Diaz Rivero et al., 2018). In this paper, we use the NFW profile to describe the mass distribution of the host dark matter halo. The NFW profile is a universal density profile that emerged from cosmological N-body simulations and describes the radial density distribution of dark matter halos as $\rho(r) \propto (r/r_s)^{-1}(1 + r/r_s)^{-2}$, where r_s is a characteristic scale radius (Navarro, 1996). An NFW halo can be fully characterised by two parameters $\{M_{200}, c_{200}\}$ (Brehmer et al., 2019). Here, M_{200} is the total mass enclosed within the virial radius r_{200} , defined as the radius within which the mean density equals 200 times the critical density of the universe. The virial mass is related to the observable velocity dispersion σ_v through an empirical scaling relation (Zahid et al., 2018):

$$\log \left(\frac{M_{200}}{10^{12} M_\odot} \right) = a + b \left(\frac{\sigma_v}{100 \text{ km s}^{-1}} \right) \quad (6)$$

where $M_\odot$ is the solar mass, $a = 0.09$, and $b = 3.48$. This relation allows us to assign a physically motivated halo mass to each simulated lens based on its velocity dispersion. Note that this relation includes an intrinsic scatter of 0.13 dex; accordingly, $\log M_{200}$ is sampled from a normal distribution centred on the above relation with standard deviation 0.13 dex. The second NFW parameter, the concentration $c_{200} = r_{200}/r_s$, describes how centrally concentrated the mass distribution is. Higher concentrations correspond to more centrally peaked density profiles. The concentration is related to the halo mass through a fitting formula calibrated on N-body simulations (Sánchez-Conde and Prada, 2014):

$$c_{200} = \sum_{i=0}^5 c_i \times \left[\ln \left(\frac{M_{200}}{0.7 M_\odot} \right) \right]^i \quad (7)$$

where $c_i = [37.5153, -1.5093, 1.636 \cdot 10^{-2}, 3.66 \cdot 10^{-4}, -2.89237 \cdot 10^{-5}, 5.32 \cdot 10^{-7}]$. This mass–concentration relation reflects the assembly history of dark matter halos: lower-mass halos formed earlier when the universe was denser, leading to higher concentrations.

The total mass of the lensing system comprises the mass of the host halo and the mass of its subhalos. While both can in principle be described by NFW profiles, the Λ CDM model predicts a large number of small-mass subhalos orbiting within the host halo. These subhalos are subject to tidal stripping by the host potential, which removes mass from their outer regions. To account for this, we use a truncated Navarro–Frenk–White (TNFW) profile (Diaz Rivero et al., 2018) for each subhalo, which smoothly truncates the density profile at a tidal radius, effectively discarding the unbound outer mass.

The number and mass distribution of subhalos within a host halo is described by the subhalo mass function. This is one of the central quantities of interest in our work, as it encodes information about the nature of dark matter. Given a host halo mass M_{200} , the differential number of subhalos as a function of subhalo mass m_{200} follows a power-law distribution (Brehmer et al., 2019):

$$\frac{dn}{d \log \frac{m_{200}}{m_{200,0}}} = \begin{cases} \alpha \frac{M_{200}}{M_{200,0}} \left(\frac{m_{200}}{m_{200,0}} \right)^\beta, & (m_{200}^{\min} \leq m_{200} \leq m_{200}^{\max}) \\ 0, & \text{else} \end{cases} \quad (8)$$

where α is the normalisation factor representing the overall abundance of substructure, and $\beta < 0$ is the power-law slope. The slope β is a physically meaningful parameter: a more negative value of β indicates that the population is dominated by low-mass subhalos, as predicted by CDM, whereas a less negative slope would imply relatively more massive subhalos. The normalisation α can be derived from the other two parameters $\{f_{\text{sub}}, \beta\}$ and the host halo mass M_{200} , through the definition of the substructure mass fraction (Brehmer et al., 2019):

$$f_{\text{sub}} = \frac{\int_{m_{200,\min}}^{m_{200,\max}} dm_{200} \, m_{200} \frac{dn}{dm_{200}}}{M_{200}} \quad (9)$$

where $m_{200,\min} = 10^7 M_\odot$ and $m_{200,\max} = 0.01 M_{200}$. The substructure mass fraction f_{sub} quantifies the total fraction of the host halo mass that resides in subhalos: a higher f_{sub} means more total mass in substructure and correspondingly more subhalos visible in the lensing image. Together, $\{f_{\text{sub}}, \beta\}$ fully specify the subhalo population for a given host halo mass, and these are the two parameters that our neural network aims to constrain.

The total number of subhalos n_{tot} within the virial radius R_{200} of the host halo can be computed by integrating the mass function: $n_{\text{tot}} = \int_{m_{200,\min}}^{m_{200,\max}} dm_{200} \frac{dn}{dm_{200}}$. However,

not all of these subhalos lie within the region probed by the lensing image. We define the region of interest as a circle of angular radius $\theta_r = 2\theta_E + 1$ arcsec, which ensures that we capture all subhalos on or near the Einstein ring where their lensing effect is most significant. The number of subhalos projected within this angular radius is computed from the NFW profile's projected mass distribution:

$$n_s(x) = n_{tot} \left[\ln \left(\frac{x}{2} \right) + F(x) \right] \quad (10)$$

where $x \equiv \theta_r/\theta_s$ is the projected angular radius in units of the angular scale radius θ_s of the host halo, and $F(x)$ is an auxiliary function that arises from the analytic projection of the NFW profile along the line of sight:

$$F(x) = \begin{cases} \frac{1}{\sqrt{x^2 - 1}} \tan^{-1} \sqrt{x^2 - 1}, & x > 1 \\ \frac{1}{\sqrt{1 - x^2}} \tanh^{-1} \sqrt{1 - x^2}, & x < 1 \\ 1, & x = 0 \end{cases} \quad (11)$$

This formulation ensures that the number of subhalos placed in each simulated image is physically consistent with the NFW density profile of the host halo, accounting for the fact that the projected density is higher near the centre of the halo.

Our aim is to predict $\{f_{sub}, \beta\}$ using a deep neural network. In the training set, f_{sub} is chosen from the uniform distribution $U[0.001, 0.2]$ and β is chosen from the uniform distribution $U[-1.5, -0.5]$ according to Brehmer et al. (2019). To test the performance of neural network with the Euclid simulated image more authentic, we implement the experiment with Gaussian point spread function (PSF) with FWHM 0.16 arcsec and shot noise. Noise from both the electronic equipment and lens depends on the exposure time (Hoffmann et al., 2014). The specified exposure time is given as 565s in Euclid telescope (Cropper et al., 2014), however, such short exposure time give the high uncertainty for network predicting. In this case, we test the effect of noise for different exposure times on model predictions.

Simulated lensing images in this paper are single-band for applying CNN. Figure 1 shows samples of simulated images for different parameter settings. The resolution of the simulated image is 0.101 arcseconds per pixel, which matches the resolution of the Euclid telescope (Cropper et al., 2014). Furthermore, we also test different resolutions of input images for the neural networks which has best performance. Except for the experiment of different resolution of input images, the input images in this paper are 60×60 pixels with 0.101" per pixel

To test and evaluate the different networks' performance, we use the simulated lensing images generated by the simplest parameter setting, which is, SIS profile with no randomness (z_{lens} fixed as $10^{-0.25}$, source location fixed as the center of image), and

no noise (not including shot noise and PSF). We choose $\{f_{sub}, \beta\} = \{0.05, -0.9\}$ as label to test following Brehmer et al. (2019).

3.2 Direct Regression Method

Deep learning networks contain interconnected layers with multiple nodes. The information is passed through the nodes of the different layers and updated by the weights connecting the different nodes. The outputs of the nodes will all be derived from the non-linear activation. As a subset of deep neural networks, CNNs use convolutional filters rather than individual nodes to extract features from each layer. Features of input image in the convolutional neural network will be extracted through the convolution kernel pixel features and non-linear activation functions are used to obtain feature maps (Albawi et al., 2017). Information from the feature maps is fed as a vector through a non-linear activation function and a fully connected layer to output the prediction. Accurate estimation can be achieved by reducing the loss between the true and predicted labels through back propagation. We used mean absolute error (MAE) comparing the difference between the prediction value $\{f_{sub}, \beta\}_{pred}$ and $\{f_{sub}, \beta\}_{true}$. All labels in the training set are normalized. And for labels in test set images, we set as $f_{sub} = 0.05$, $\beta = -0.9$.

As input to the CNN, single-band images of 60×60 pixel size with $0.101''$ per pixel and 120×120 pixel size with $0.0505''$ per pixel were used as input images for training and testing, and all images are normalised. The CNN structure for 60×60 input image is shown in Figure 2. EfficientNet has the ability to adjust the structure of networks using given network parameters of the resolution scale s , depth d , and width w . We compare the ratio of the three parameters from EfficientNetB0 to EfficientNetB7 (Tan and Le, 2019) and set $d = 0.23$ and $w = 0.358$ for 60×60 input images and $d = 0.6$ and $w = 0.6$ for 120×120 input images. Between the connection of convolutional layer, we applied batch normalisation layer and Rectified Linear Unit (ReLU) (Nair and Hinton, 2010). Batch normalisation is used to ensure that the iterative learning process has the same data distribution and ReLU act as non-linear function.

In network structures, besides using EfficientNet for the CNN, the deep neural network requires fully connected layers to receive the feature map extracted by EfficientNet and two neurons for prediction. In the middle of EfficientNet and fully connected layers, we add a flatten layer to convert the extracted feature image into a one-dimensional vector. We use 80,000 images used for training, 20,000 images used for validation, and 1000 used for test. For hyperparameter settings, lensing images are trained with a batch size of 32 and learning rate of 0.001. We use a novel optimiser, Adabelief, which adapts the step size according to the gradient direction and outperforms other optimisers in some fields of image processing (Zhuang et al., 2020).

3.3 Simulation-Based Inference Method

As introduced in Section 2.4, the goal of Bayesian inference is to compute the posterior distribution of the subhalo population parameters given an observed lensing image. Applying Bayes’ theorem to our problem:

$$p(\theta | x) = \frac{p(x | \theta) \cdot p(\theta)}{p(x)} \quad (12)$$

where $\theta = \{f_{\text{sub}}, \beta\}$ are the subhalo population parameters we wish to constrain, and x denotes a simulated lensing image. The prior $p(\theta)$ is set as a uniform distribution over the physically motivated range specified in Section 3.1: $f_{\text{sub}} \in [0.001, 0.2]$ and $\beta \in [-1.5, -0.5]$. The posterior $p(\theta | x)$ represents the probability of different parameter values given the observed image—it is the quantity we ultimately want to estimate. However, as discussed in Section 2.4, the likelihood $p(x | \theta)$ and evidence $p(x)$ are both analytically intractable for our image simulation pipeline, making it impossible to compute the posterior directly.

To circumvent this intractability, we employ Sequential Neural Ratio Estimation (SNRE), a simulation-based inference method that uses a neural network classifier to approximate the likelihood-to-evidence ratio (Durkan et al., 2020; Hermans et al., 2020). The key insight of SNRE is that rather than estimating the likelihood and evidence separately, one can train a classifier to directly learn their ratio, which is sufficient for posterior inference. Through iterative rounds of training, the posterior estimated in the previous round can optionally be used as the proposal distribution for the next round, progressively focusing the training on regions of high posterior density (Durkan et al., 2020).

The method works by framing posterior estimation as a classification problem. Given two different parameter vectors θ_0 and θ_1 , the neural network is trained as a binary classifier $\widetilde{f}_\varphi(\theta, x)$ that distinguishes which parameter generated a given image. The optimal classifier output is related to the log-ratio of the likelihood to the evidence:

$$\widetilde{f}_\varphi(\theta, x) = \log \frac{p(x | \theta)}{p(x)} + c(x) \quad (13)$$

where $c(x)$ is a constant with respect to θ that cancels in ratio computations, and φ denotes the learnable parameters of the neural network. This relationship means that by training a sufficiently expressive classifier, we obtain an approximation to the likelihood-to-evidence ratio. The classifier output can be used to express the posterior probability of θ_1 versus θ_0 through a softmax:

$$p(\theta_1 | x) = \frac{\exp(\widetilde{f}_\varphi(\theta_1, x))}{\exp(\widetilde{f}_\varphi(\theta_0, x)) + \exp(\widetilde{f}_\varphi(\theta_1, x))} \quad (14)$$

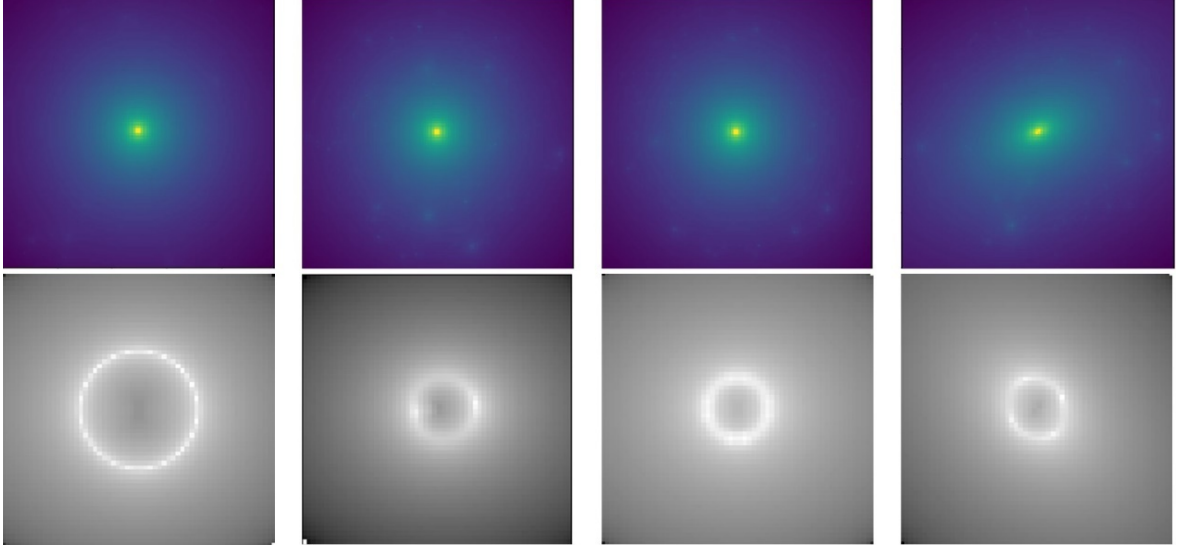


Figure 1: Sample of simulated gravitational lensing images. 5 images (upper panels) are simulated lensing images contain one host halo and different subhalos and 5 images (lower panels) are of single-band images that are used for training and contain the Einstein ring in the image above. Parameter settings of the simulated lensing images, from left to right: SIS profile with simplest parameter, SIS profile and random redshift and random source galaxy, SIS profile and PSF and shot noise, and SIE profile. All sample images have dimensions of 60×60 pixels and a resolution of $0.101''$ per pixel and are generated using parameters f_{sub} and β fixed at 0.05 and -0.9 respectively.

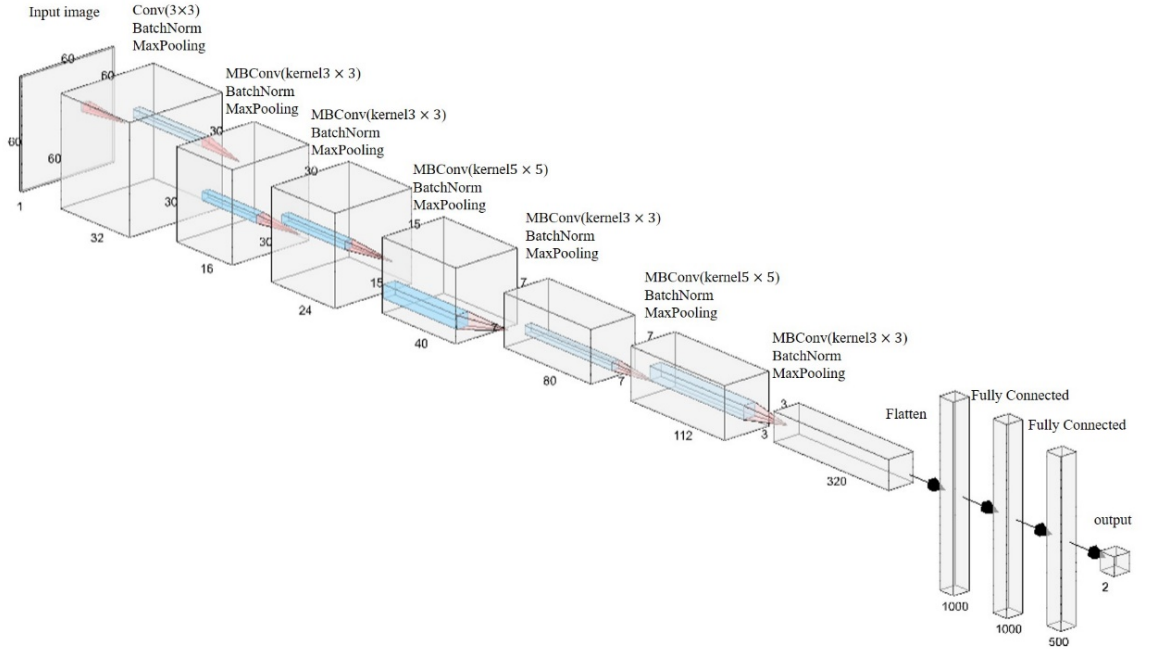


Figure 2: Structure of EfficientNet and fully connected layers used in Section 3.2. MBConv is also referred to as an inverted residual block. Numbers are the height, width, and depth for each stage.

The binary classification framework generalises naturally to a multi-class setting known as “contrastive learning with atoms.” Instead of distinguishing between just two parameter vectors, the network learns to identify which of K candidate parameters actually generated a given image. This is formalised through the following cross-entropy loss function (Durkan et al., 2020):

$$\mathcal{L}(\varphi) = -\frac{1}{B} \sum_{b=1}^B \log \frac{\exp(\widetilde{f}_{\varphi}(\theta^{(b)}, x^{(b)}))}{\sum_{k=1}^K \exp(\widetilde{f}_{\varphi}(\theta^{(k)}, x^{(k)}))} \quad (15)$$

where B is the batch size (number of images per training step), K is the number of “atomic” parameter proposals per image (with $0 < K - 1 < B$), and φ denotes the neural network weights that are optimised via stochastic gradient descent. The loss has a clear interpretation: it measures how well the classifier can identify the correct parameter–image pair among K candidates. Note that the loss is a negative number (bounded above by zero), so values closer to zero indicate better classification performance and, consequently, a more accurate approximation to the likelihood-to-evidence ratio.

Once the classifier is trained, it provides an approximation $\widetilde{f}_{\varphi}(\theta, x) \approx \log[p(x | \theta)/p(x)]$, from which the posterior can be evaluated up to a normalisation constant. However, computing the normalised posterior $p(\theta | x)$ requires integrating over the full parameter space, which is not analytically feasible given the complexity of the learned function.

MCMC Sampling. To obtain posterior samples without computing this integral, we use Markov Chain Monte Carlo (MCMC) sampling with the learned log-ratio as the unnormalised log-posterior. Specifically, we employ slice sampling for adjusting its step size to efficiently explore the posterior landscape without requiring manual tuning of a proposal distribution. We use the default MCMC configuration of the `sbi` library for the number of chains, warmup steps, and thinning factor. For each test image, we draw a total of 5,000 MCMC samples, which we verified is sufficient for the posterior contours to stabilise.

To improve the quality of the posterior approximation, we use 20 atomic proposals per batch element during training. Each atomic proposal represents a different contrastive parameter vector sampled from the prior; using more atoms provides the classifier with a richer set of comparisons at each training step, resulting in a more accurate approximation to the likelihood-to-evidence ratio (Greenberg et al., 2019).

Because the subhalo population parameters $\{f_{\text{sub}}, \beta\}$ are shared across all lensing images drawn from the same cosmological model, the inference is performed at the population level. For N test images $\{x_1, x_2, \dots, x_N\}$ that are independent and identically distributed (i.i.d.) given the population parameters, the joint posterior factorises as $p(\theta | x_1, \dots, x_N) \propto p(\theta) \prod_{i=1}^N r(x_i, \theta)$. In our experiments, we use $N = 1,000$ test images,

and the product of ratios is computed in log-space for numerical stability.

In practice, the SRE training procedure proceeds as follows. At each training step, a batch of B jointly distributed pairs $\{(\theta^{(b)}, x^{(b)})\}_{b=1}^B$ is drawn from the joint distribution $p(\theta, x)$. For each pair, $K - 1$ “contrastive” parameters are sampled from the marginal prior $p(\theta)$ to form the set of K atoms. The classifier is trained to identify which of the K candidate parameters actually generated the observed image, using the cross-entropy loss of Equation 15. This contrastive mechanism is critical: it forces the network to learn the likelihood-to-evidence ratio rather than merely memorising training examples. We use $K = 20$ atomic proposals per batch element. After each epoch, the model is evaluated on a held-out validation set comprising 20% of the simulated data. Training proceeds for a maximum of 400 epochs.

The neural network structure we used containing the Network Embedding (NE), which used to combine two networks together. The Network Embedding technology allows to map the high-dimensional vectors to low-dimensional space and use the low-dimensional vectors as input for other machine learning models. The former neural network is CNN are used as NE to extract lensing image features and output a feature vector which can be learned in the next neural network. In the training process of deep neural network, the error gradient decreases with backpropagation in the chain rule. Therefore, the initial layers of the deep structure will be updated slowly even not updated because the gradient vanishes. In order to solve it and allow the later layers to also learn initial layers information, we applied the ResNet-like network in latter neural network, which contains residual blocks to maintain the gradient during training (He et al., 2016), and the latter neural network used to train and classify the posterior based on given images and prior. Conceptually, the embedding network replaces the need for hand-designed summary statistics: by training end-to-end with the ratio estimation loss, the CNN automatically discovers a compressed image representation that preserves information about $\{f_{\text{sub}}, \beta\}$ while discarding irrelevant variation. This data-driven compression is especially valuable when the mapping from parameters to observables is complex, as is the case for substructure lensing. We note that batch normalisation is omitted from the classifier portion of the network, following evidence that it can distort estimated posteriors in likelihood-free settings (Hermans et al., 2020).

The former network is a simple CNN with only has a few convolutional layers to extract features of training images and we called SimpleNet here. SimpleNet maps the extracted features to a $1 \times 1 \times 100$ vector. This vector is concatenated with θ to a $1 \times 1 \times 102$ vector and input into the ResNet-like network. We also tested using different CNN (e.g. EfficientNet, ResNet) as NE, but the results of estimates are all unacceptable and training speed is too long. We show a result of comparison of using EfficientNet as NE and using SimpleNet as NE to output $1 \times 1 \times 100$ vector in Section 4.2. The latter network or ResNet-like neural network we use contains 8 layers with two sets of residual

blocks. The residual block consists of two 50×1 dense (fully connected) layers and skip connection to maintain the gradient. The $1 \times 1 \times 102$ vector composed of x and θ is used as the input of ResNet to output the max approximation classifier $\widetilde{f}_{\varnothing}(\theta, x)$. The deep neural networks optimise the loss function in Equation 15 iteratively.

For hyperparameter settings, we set 0.0005 as learning rate, a training batch size of 100, and a validation fraction of 0.2. The max number of epochs is 400. We also apply the data augmentation in this section. During training process, the training set images are randomly rotated in 90 degrees increments.

Generally, one measure used in evaluating the performance of the network structure is the decrease in the difference between the predicted label and true label as the number of training steps or epochs increases. However, this measure does not apply to the simulation-based inference training process. The relation between the epochs and error during training is shown in Figure 3. For the convenience of explanation, we do not use early stopping or the training process. The results in Figure 3 can be generalised to all experiments when using SimpleNet for NE in the SBI neural network. We use 10 MCMC samples to calculate the MAE, which is the average distance from the predicted sample point to the true label. The figure shows that there is no evidence of a linear relationship between the epoch number and MAE although the MAE has a downward trend before approximately epoch 70. Starting from epoch 200, the MAE oscillates at approximately 0.1 but does not converge, and we thus set the maximum number of epochs at 400. Additionally, a relatively low value of MAE appears largely in areas with high validation loss, but there is again no linear relationship. This shows that the deep neural network with SBI cannot directly use the difference between the true label and predicted label as a loss function to optimise the weight through back propagation. Therefore, the deep neural network uses Equation 15 as the loss function to find the accurate estimates of predicted sample points and true labels.

In addition to the experiment of different resolution of input image, the SBI network structure using SimpleNet for NE shown in Figure 4. Input images with different dimensions require different convolutional network depths. We therefore remove the first convolutional layer of SimpleNet for 30×30 input images and add a convolutional layer that is the same as the last convolutional layer after the last layer for 120×120 input images.

4 Results

This section presents the results of the above implementation. In following subsections, we show the results of predicting the mass parameters of substructure using only EfficientNet in Section 4.1. Section 4.2 shows the results of using SimpleNet for NE in SBI, and using EfficientNet for NE in SBI. Section 4.3 presents the results of training

and testing simulated

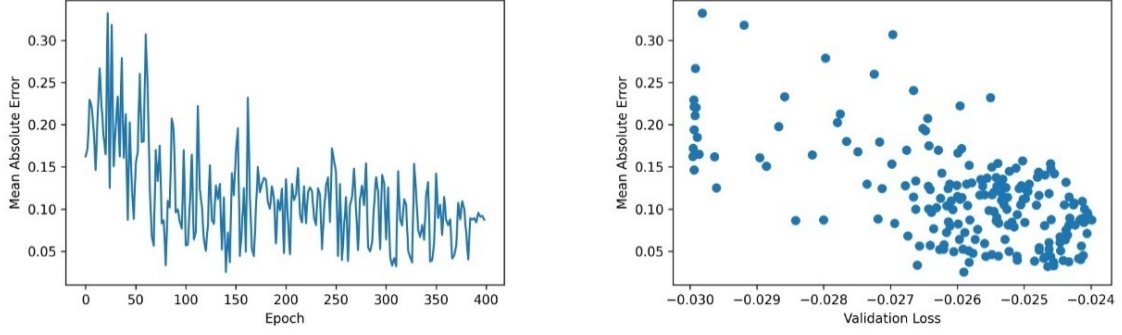


Figure 3: Relationships between the MAE and the epoch (left) and validation loss (right).

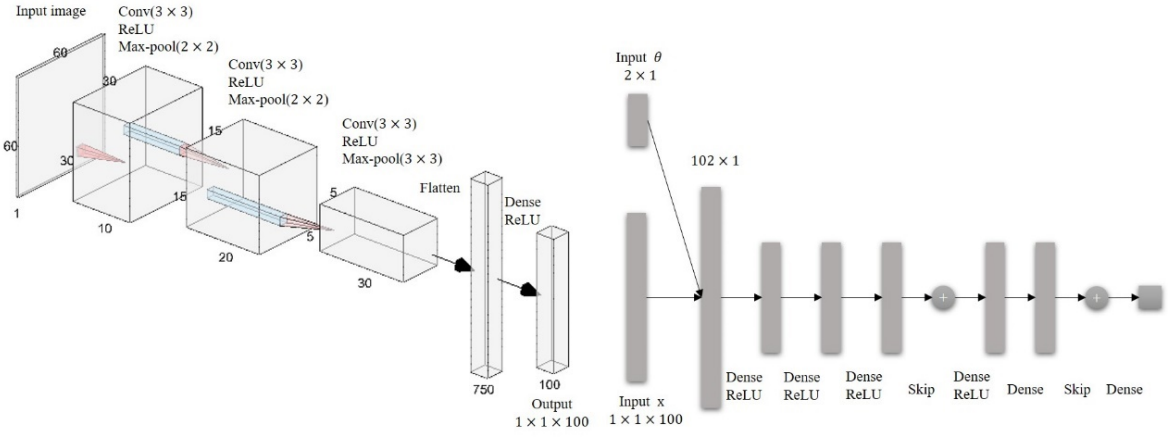


Figure 4: Structure of SimpleNet and ResNet-like used in simulated-based inference. There are 13 layers in total. The left image shows the structure of SimpleNet and the right image shows the structure of the ResNet-like neural network. θ is a vector of parameters $\{f_{sub}, \beta\}$. The output in the last layer of SimpleNet is the input x of the ResNet-like network.

images generated by more complex parameters using SBI. Again, because the loss is negative in SBI, the accuracy increases as the loss rises. The validation loss in the SBI section is an important result used to evaluate posterior estimates. The highest validation loss determines the posterior used in testing.

4.1 Direct Regression Results

After generating simulated images according to Section 3.1, 80,000 pictures are used for the training set and 20,000 pictures are used for the validation set. Both the training set and validation set are generated by selecting values from the uniform distribution $\{f_{sub}, \beta\}$. The lensing image used for testing generated for $f_{sub} = 0.05$, $\beta = -0.9$. Figure 5 is a scatter plot of the predictions for two sizes of input image, illustrating the effect of the resolution of the input image on the prediction. The average MAE

is calculated as 0.085 ± 0.0004 for the 60×60 input image and 0.124 ± 0.0002 for the 120×120 input image. In adopting EfficientNet-only to train and test single-band images simulated using the simplest parameters (SIS profile without randomness or noise), a larger input image reduces the prediction accuracy. This suggests that using CNN alone is not sufficient to obtain accurate estimates. Moreover, using EfficientNet to train the simulated images with the SIE profile or any nuisance parameter, the validation loss fails to converge and cannot give accurate prediction so we do not show the results here. This failure is not surprising: as discussed in Section 2.1, substructure perturbations affect only a small number of pixels in the Einstein ring, and a standard regression CNN has no mechanism to distinguish these subtle perturbations from noise or macro-model variations. The network instead learns to predict the mean of the training label distribution, which minimises the MAE but provides no useful information about individual images. These results motivate the use of simulation-based inference, which frames the problem as density estimation rather than point prediction.

4.2 SBI Neural Network with Different Network Embedding

In applying an SBI neural network, we use CNN to extract image features for NE. We here apply the same dataset used in Section 4.1. Figure 6 shows the kernel density estimation (KDE) of the difference between estimates and true labels. The kernel density estimation is an estimate of the probability distribution of the sample when the overall sample is unknown. The probability density is greater when the sample points are denser. Kernel density estimates are well suited to visualising SBI predicted results since estimates are all sampled by MCMC. Figure 6 shows the difference of estimates and true label $\{0.05, -0.9\}$. Figure 6 shows using EfficientNet for NE, the mean value of difference has huge bias compared with when using SimpleNet for NE the mean value of difference is close to 0. The best validation loss is -0.046 ± 0.0008 when using EfficientNet for NE and -0.020 ± 0.0010 when using SimpleNet for NE. The figure shows that using SimpleNet for NE gives an acceptable range irrespective of f_{sub} or β in contrast with the case of using EfficientNet for NE. The average MAE is calculated as 0.005 ± 0.0001 when using SimpleNet for NE and 0.160 ± 0.0010 when using EfficientNet for NE. The substantially better performance of SimpleNet is noteworthy given that EfficientNet is a more complex architecture with a larger number of parameters. We attribute this to the fact that a simpler embedding network produces a more stable gradient signal for the downstream ratio estimation classifier: EfficientNet’s many layers and compound scaling introduce additional optimisation difficulties when the embedding is trained end-to-end with the SBI loss, which is fundamentally different from the standard classification or regression losses for which EfficientNet was designed. This finding suggests that for SBI applications, the choice of embedding network should

prioritise training stability over raw representational capacity.

4.3 SBI Performance with Nuisance Parameters

This section presents the SBI neural network performance for simulated lensing images generated with different nuisance parameters. The results of Section 4.2 showed that it is better to use SimpleNet rather than EfficientNet for NE in SBI, and we therefore implement only the former in experiments on simulated images involving randomness, electronic instrument noise with different exposure times, and different ellipticity. The simulated images have dimensions of 60×60 pixels and resolution of $0.101''$ per pixel to match the Euclid configuration (Cropper et al., 2014), but we also show the effects of the resolution of input images on the prediction. In order to test the performance of SBI neural network in our paper, we tested different $\{f_{sub}, \beta\}$ combination in test results. According to Diaz Rivero and Dvorkin (2020), for subhalos range from 10^6 to $10^{11} M_\odot$, the bound of f_{sub} is 0.01. Additionally, we added $\beta = -1.4$ and $f_{sub} = 0.1$ in to test set. Note that low f_{sub} value indicate that the number of subhalos is less distributed in the lensing image while low β values indicate that the proportion of small mass subhalos is higher. We therefore set $\theta^i = \{0.01, -1.4\}, \{0.01, -0.9\}, \{0.05, -1.4\}, \{0.05, -0.9\}, \{0.1, -1.4\}, \{0.1, -0.9\}$.

The results of the previous experiments are used to decide which deep neural network is suitable for estimating the subhalos mass parameters and lensing images are generated by relatively simple parameters. However, galaxy–galaxy lensing images are not static, and we thus add nuisance parameters to make the simulated image more realistic. In this section, we use the best validation loss to evaluate the performance of the SBI neural network and confidence intervals of 68.27% and 95.45% to evaluate the accuracy of the posterior or

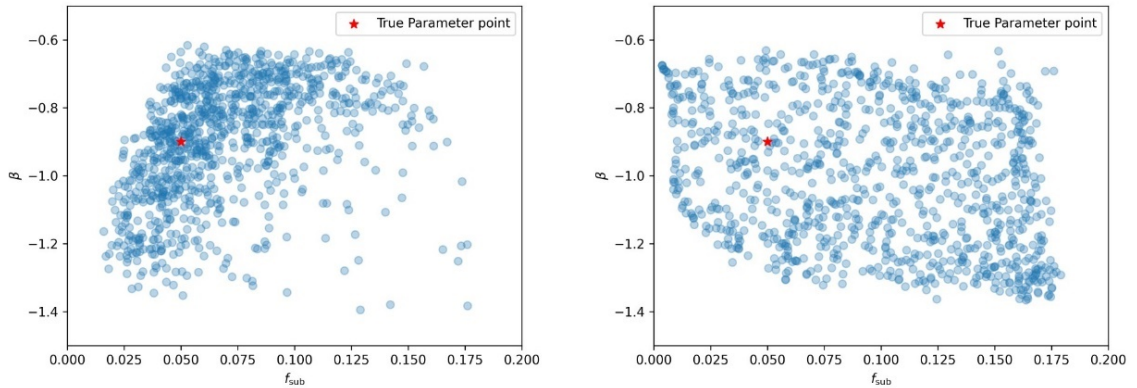


Figure 5: Scatter plot of 1000 estimates when using EfficientNet to predict $\{f_{sub}, \beta\}$. The left image is the prediction for an input image having dimensions of 60×60 pixels and resolution of $0.101''$ per pixel whereas the right image is the prediction for an input image having dimensions of 120×120 pixels and resolution of $0.0505''$ per pixel.

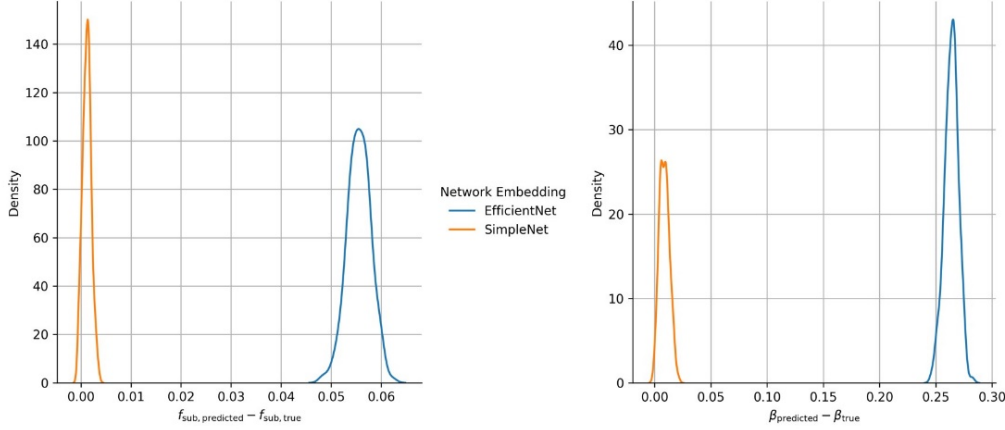


Figure 6: Comparison of the difference of the network-predicted performance of using SimpleNet and EfficientNet for NE. The orange line and blue line are kernel density estimates of the difference between predicted and true values when using SimpleNet for NE and when using EfficientNet for NE respectively. The left image shows difference of $f_{sub,predicted}$ and $f_{sub,true}$. The right image is difference of $\beta_{predicted}$ and β_{true} .

estimate samples obtained using MCMC. The narrower the confidence interval, the higher the estimation accuracy. It is worth noting that the posteriors sampled by MCMC do not cover all posterior regions; therefore, the confidence intervals may be discontinuous in some cases.

The confidence interval of estimates after adding randomness is shown in Figure 7. As the simulated lensing images contain random lens redshifts and source galaxy locations to be trained and tested, the network will learn from different lensing light and host locations in the lensing images. Although the addition of randomness will slightly reduce the accuracy of estimators, it provides robustness for multiple predictions. Figure 7 shows that in the case of the lens image with added randomness, the confidence interval of the estimator is obviously widened, but at same time, for the parameter points $\{0.01, -0.9\}$, the estimator gives an acceptable range. The best validation loss is -0.022 ± 0.003 for the data set with added randomness and -0.020 ± 0.001 for the data set with the simplest parameters, which provides an acceptable drop in validation loss.

In the experiments of adding noise with different exposure times, we also add randomness because it provides an acceptable accuracy and validation loss. The noise parameters of the PSF and shot noise relate to the electronic instrumentation. Both noises should be considered comprehensively because the exposure time affects the shot noise and PSF of the telescope (Hoffmann et al., 2014). The image becomes clearer as the exposure time increases, and lower exposure times result in the inaccurate extraction of features by the network. The exposure time of the Euclid telescope is 565 s in general (Cropper et al., 2014). However, such a short exposure time widens the confidence interval of the estimator sample and does not cover the true label as shown in the first graph of Figure 8. In this case, we apply a longer exposure time to find a

suitable time for the SBI network. We extend the exposure time of the Euclid telescope by factors of 2, 5 and 10. Results are presented for the network prediction in Figure 8 and the validation loss in Table 1. The results in Figure 8 show the tendency that the confidence interval for each sample of estimates becomes narrower as the exposure time increases. Note that when the exposure time is 2825s, although the confidence interval for the predicted $f_{sub} = 0.01$ covers the true label, the confidence interval for the same colour appears at a different position, indicating that the posteriors sampled by MCMC appear in different regions and that the posterior probabilities are highly biased. The validation loss in Table 1 illustrates that increasing the exposure time helps in predicting the accuracy of the model.

In the spirit of stacking multiple observations, we next conduct an experiment on input images having different pixel size and resolution of input images. In this experiment, we select all acceptable nuisance parameters into test. The lensing images are simulated by SIS profile, random z_{lens} and source location, PSF and shot noise with 5850s exposure time. We tested the input images with 30×30 , 60×60 and 120×120 pixels for comparing. The confidence interval of estimates for input images with 30×30 and 120×120 pixels are shown in Figure 9 and the confidence interval of estimates for input images with 60×60 pixels is the result for the exposure time of 5850 s in Figure 8. For input images having dimensions other than 60×60 pixels, we change the CNN structure as mentioned at the end of Section 3.3. Figure 9 shows that the resolution of the input image affects the range of confidence intervals, with larger resolution resulting in narrower confidence intervals and Table 1 shows that greater resolution of the input image effectively reduces the absolute value of the validation loss. The absolute value of the validation loss decreases 0.001 ± 0.0002 when scaling pixel values in multiples. It is clear that when the network trains input images with high pixel values, it increases the accuracy of the predicted parameters, but this does not help in predicting parameters with $f_{sub} = 0.01$.

In the experiment of SIE profile, we test the SIE profile with different ellipticity by multiplying the standard deviation and degree of the ellipticity parameters with factor σ_m . Figure 10 presents the confidence interval of estimates for the gravitational lens mass obtained with the SIE profile for different degrees of ellipticity. The result of $\sigma_m = 1$ shows great uncertainty in the prediction $\{f_{sub}, \beta\}$, which is the result of using the specified ellipticity parameter. However, as the standard deviation and max degree of orientation decreases, the confidence interval of the prediction narrower, which shows that the deviations and degree of ellipticity greatly affect the accuracy of network predictions. It is worth noting that the confidence interval for the predicted f_{sub} becomes progressively narrower as the deviation from the ellipticity parameter and the maximum orientation angle decrease. However, the confidence interval of the predicted $f_{sub} = 0.01$ still does not cover the true label due to the smaller number of subhalos indicated by

the relatively small f_{sub} . Additionally, there are two confidence interval contours for the prediction $\{0.1, -1.4\}$ in the results of $\sigma_m = 0.5$, indicating that the sample by MCMC for the test set posterior does not cover the full region of posterior. $\sigma_m = 0$ indicate using SIE profile and full orientation degree but parameters without deviation and fixed with mean value for ellipticity parameters. The confidence interval of estimates shows an acceptable range except for the case of $f_{sub} = 0.01$. Table 1 shows the effect of different ellipticities on the validation loss. The absolute value of the validation loss rises as the standard deviation of the ellipticity parameter increases and as the maximum orientation angle decreases.

5 Discussion

Strong gravitational lensing is a powerful tool to probe and quantify the dark matter mass on sub-galactic scales. However, such lensing images are difficult to obtain in large quantities. In this paper, we used Lenstronomy (Birrer and Amara, 2018) to simulate one-band Euclid images for training and testing with required parameters. Strong galaxy–galaxy lenses contain a host halo of dark matter and smaller subhalos of dark matter. The mass distribution of these subhalos affects the gravitational lensing images and the mass model of dark matter can be parameterised by $\{f_{sub}, \beta\}$. On this basis, we applied deep neural network methods to process lensing images and predict the fraction of the contained subhalo mass f_{sub} and the slope of the subhalo mass function β . In all experiments reported in this paper, we used 80,000 lensing images for training, 20,000 lensing images for validation, and 1000 lensing images for testing.

5.1 Direct Regression Approach

CNNs are widely used for image feature extraction, and we applied EfficientNet for this purpose due to its strong performance in object recognition and its ability to optimize network structure through compound scaling of depth and width (Tan and Le, 2019). However, unlike macro-model parameters such as the Einstein radius, which directly affect the spatial distribution of pixel intensities in a lensing image, the substructure parameters $\{f_{sub}, \beta\}$ influence only subtle, localized perturbations in the Einstein ring. This makes it inherently difficult for a standard CNN to learn a mapping from image pixels to $\{f_{sub}, \beta\}$ through back propagation alone.

Figure 5 confirms this limitation: although some individual estimates fall near the true labels, the overall scatter is large. Notably, increasing the input image resolution from 60×60 to 120×120 pixels worsens the MAE from 0.085 ± 0.0004 to 0.124 ± 0.0002 . We attribute this counter-intuitive result to the fact that higher-resolution images contain more pixels that are unrelated to substructure perturbations, effectively increasing the

noise-to-signal ratio for the CNN without a corresponding increase in network capacity. We also experimented with different hyperparameters (learning rate, training step size, fully connected layer depth), data augmentation, and alternative CNN architectures including MobileNetV2 (Sandler et al., 2019) and ResNet (He et al., 2016), but none of these yielded meaningful improvements,

Table 1: Validation loss of the SBI neural network under different nuisance parameter configurations.

Nuisance parameter	Setting	Validation loss
<i>(a) Exposure time (SIS + PSF + shot noise)</i>		
	565 s	-0.027 ± 0.0006
	1130 s	-0.027 ± 0.0002
	2825 s	-0.026 ± 0.0005
	5850 s	-0.026 ± 0.0001
<i>(b) Input resolution (SIS + randomness + noise, 5850 s)</i>		
	30×30 px ($0.202''/\text{px}$)	-0.027 ± 0.0002
	60×60 px ($0.101''/\text{px}$)	-0.026 ± 0.0001
	120×120 px ($0.0505''/\text{px}$)	-0.025 ± 0.0004
<i>(c) SIE ellipticity (SIE + randomness)</i>		
	$\sigma_m = 0$	-0.023 ± 0.0004
	$\sigma_m = 0.25$	-0.026 ± 0.0005
	$\sigma_m = 0.5$	-0.027 ± 0.0003
	$\sigma_m = 1$	-0.029 ± 0.0001

despite the lensing images being generated with minimal complexity. We infer the regular CNN methods are not suitable for this case.

5.2 Simulation-Based Inference

We then applied a novel technique referred to as simulation-based inference (Cranmer et al., 2020; Brehmer et al., 2019; Hermans et al., 2020), which is based on Bayesian theory. Similarly, SBI technology also requires CNN to extract features as back end but it is trained to learn and classify the posterior probability $p(\theta|x)$ priori $p(\theta)$ and simulated images x , where θ is $\{f_{sub}, \beta\}$. The value of the posterior probability corresponds to the predicted value of the subhalos mass parameter but is are hard to integrate due to the conditions of multiple simulated images. Hence, predicting the posterior probabilities requires the properties of Markov chains where each step depends on the properties of the previous step (Brehmer et al., 2020). An estimate of $\{f_{sub}, \beta\}$ was obtained for a sample of the posterior obtained using MCMC. Classifying the posterior requires the adoption of neural network. The SBI neural network structure that we constructed involves a CNN for NE to extract latent variables of simulated images as a vector space and a ResNet-like neural network used to optimise the posterior. We tried using

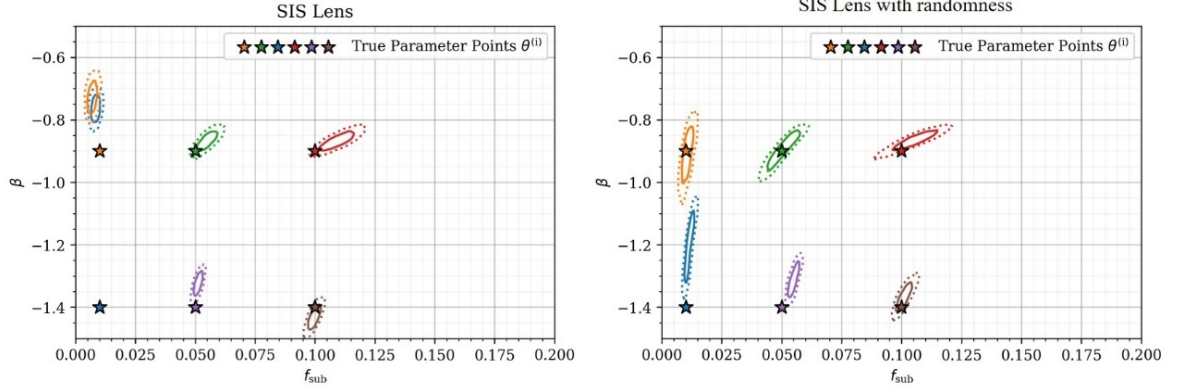


Figure 7: Comparison of the confidence intervals for network-predicted subhalo mass parameters. The left panel shows the SIS profile without nuisance parameters and the right panel shows the SIS profile with randomness. Stars mark the true label for each parameter. Solid and dashed lines are contours of the 68.27% and 95.45% confidence intervals respectively.

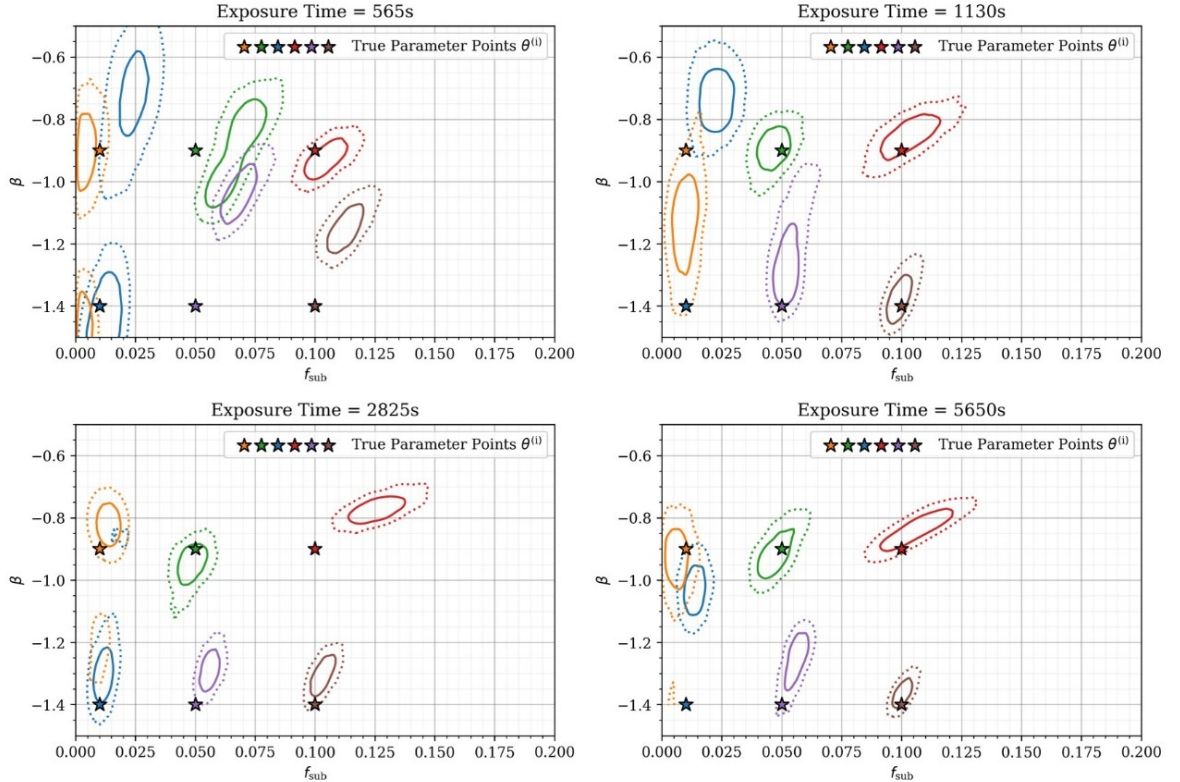


Figure 8: Comparison of the confidence intervals for network-predicted subhalo mass parameters with exposure times extended by factors of 1, 2, 5, and 10. Lensing images use the SIS profile with randomness, PSF, and shot noise. Line styles and symbols are the same as in Figure 7.

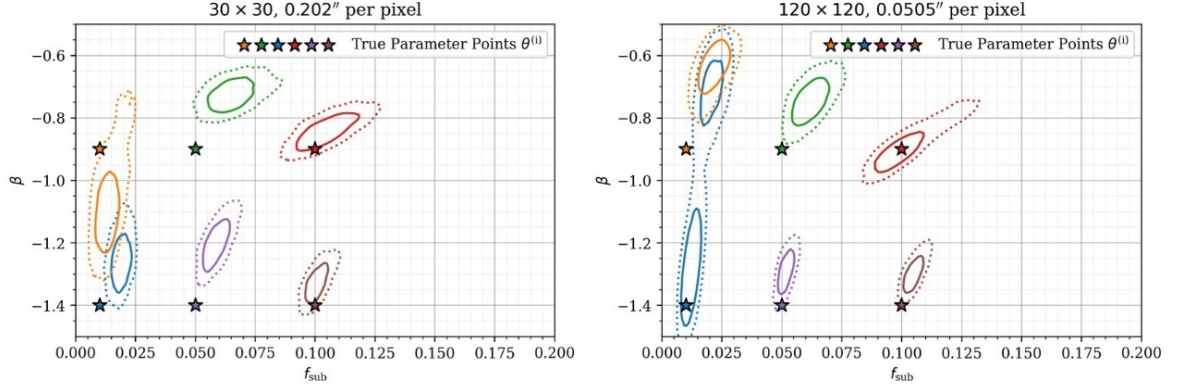


Figure 9: Comparison of the confidence intervals for network-predicted subhalo mass parameters with different input image resolutions. Lensing images use the SIS profile with randomness and noise at 5850s exposure time. Left: 30×30 pixels ($0.202''/\text{px}$); right: 120×120 pixels ($0.0505''/\text{px}$). Line styles and symbols are the same as in Figure 7.

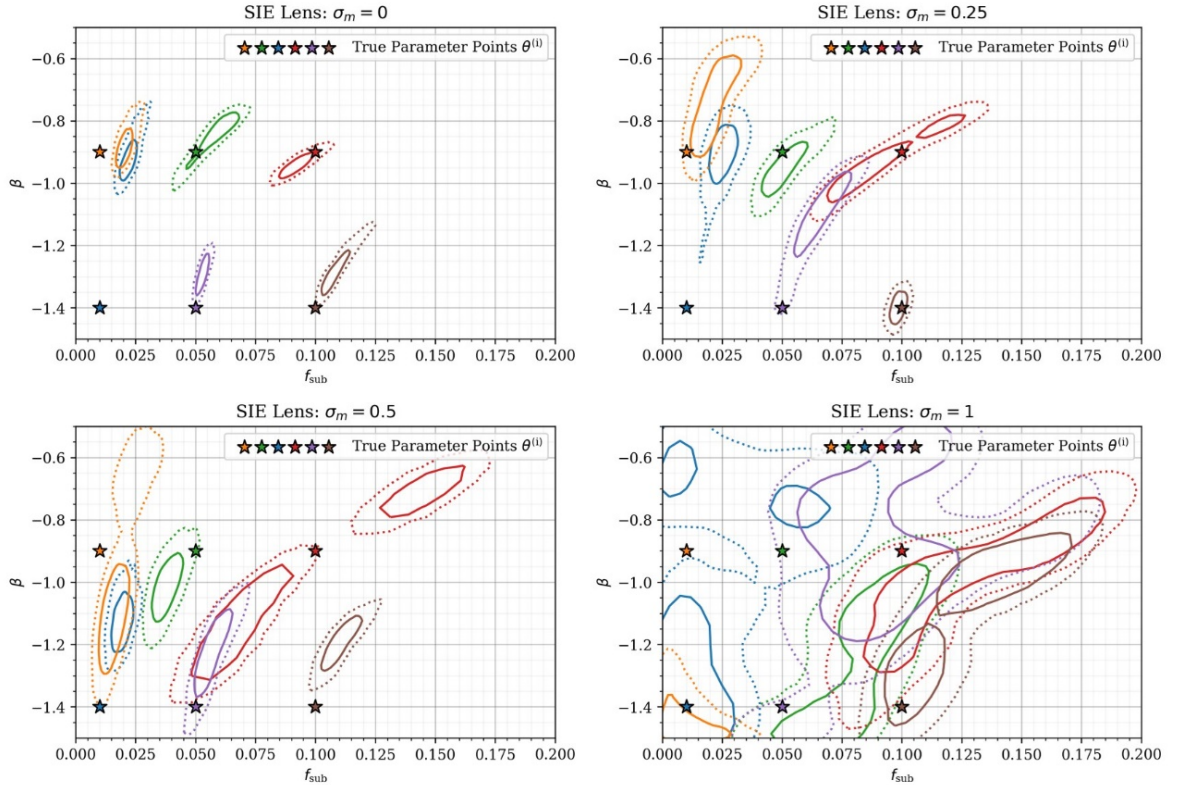


Figure 10: Comparison of the confidence intervals of network-predicted subhalo mass parameters for different degrees of ellipticity. Lensing images use the SIE profile with randomness only. σ_m denotes the coefficient used to vary the ellipticity as described in Section 3.1. Line styles and symbols are the same as in Figure 7.

EfficientNet for NE but the results were not ideal. We therefore considered building a simple convolutional neural network with fewer layers. The results in Section 4.2 show that when using EfficientNet for NE in SBI, the best validation loss has an absolute value of 0.046 ± 0.0008 and when using SimpleNet for NE in SBI, the best validation loss has an absolute value of 0.020 ± 0.0010 , the absolute value of the validation loss decreases by 50 ± 7 per cent. The MAE is 0.005 ± 0.0001 when using SimpleNet for NE. MAE decreased by 0.080 ± 0.0003 compared to the EfficientNet-only approach and by 0.150 ± 0.0009 compared to using EfficientNet for NE. The kernel density plot in Figure 6 shows that even if the error between the true and predicted values does not reach zero when using SimpleNet for NE, the performance is far better than using EfficientNet for NE. Hence, the deep neural network that uses SimpleNet for NE in SBI had the best performance and was used in subsequent experiments.

5.3 Effects of Nuisance Parameters

On this basis, we conducted several experiments to estimate $\{f_{sub}, \beta\}$ with different parameter settings of strong lensing image, including setting of randomness, noise, the input image dimensions, and the strong lensing profile. In Section 4.3, we presented the results of network-predicted estimators when using lensing images with varying randomness, exposure times, ellipticity, and pixel size and resolution of input image. The confidence interval of the sample from the posterior obtained using MCMC and the validation loss were used as evaluation criteria. However, the confidence interval is not suitable to quantify the uncertainty in this paper since we evaluate at six different parameter points simultaneously. A large standard deviation of the parameter estimates will lead to increase in the overall standard deviation, which is misleading. The accuracy of estimates is determined from the posterior that is related to the best validation loss. We therefore quantify the best validation loss as ‘uncertainty’, and the validation loss starts to increase from -0.046 during the training stage.

The findings from adding different nuisance parameters are as follows:

(i) Adding random lens redshifts and the location of the centre of the source galaxy provides acceptable accuracy in predicting the mass parameters of subhalos. The absolute value of the best validation loss when adding randomness is 0.022 ± 0.003 , which is only 4 ± 0.3 per cent greater than that of the simplest lensing images. The width and centre of the confidence interval do not change appreciably when adding randomness. Furthermore, adding randomness positively affects the prediction of parameters with $f_{sub} = 0.01$. Since the distribution of substructure is fairly broad when the lensing image is simulated with a random lensing redshift (Robertson and Smith, 2020). For lensing images with a small proportion of substructure, the probability that a CNN can extract substructural features increases.

(ii) The experimental results of shot noise and PSF show that different exposure times have a great impact on performance of the SBI network. Lensing images obtained with an exposure time of 565 s had an absolute value of the validation loss (0.027 ± 0.0006) that was 11 ± 0.2 per cent greater than that obtained with data sets that only included randomness. The range of the confidence interval of the estimates hardly covered the true label, and all the standard deviations increased with the addition of noise. The validation loss and estimates improved in experiments conducted for a long exposure time. When the exposure time was extended by a factor of 10, to 5850 s, although the standard deviation of each confidence interval was higher than that when no noise was included, the confidence interval basically covered the true label. For validation loss, the absolute value (0.026 ± 0.0001) reduced by 2 ± 0.2 per cent when the exposure time was increased to 5850 s. Although the addition of PSF and shot noise will cause the network-predicted estimation to perform worse, it provides an acceptable range under the conditions of a long exposure time.

(iii) In the spirit of stacking multiple observations, we selected all acceptable parameters of lensing images and conducted experiments with input images having different resolutions. The simulated lensing images contained randomness, had a Gaussian PSF, and had shot noise for an exposure time of 5850 s in the experiment on different resolutions of the input image. The absolute value of the validation loss decreased as the resolution of the input images increased as shown in Table 1. The multiple scaling of the input image resulted in a 2 ± 0.2 per cent change in the validation loss. There was slight improvement when using images having resolution of 120×120 pixels with 0.0505 arcseconds per pixel in terms of the confidence intervals presented in Figures 8 and 9. Although larger pixel size and higher resolution of the input image resulted in lower absolute values of the validation loss, the SBI neural network performed differently in predicting different mass parameters of subhalos. For the predicted subhalos mass parameters of $\{0.05, -1.4\}$ and $\{0.1, -0.9\}$, the confidence interval of estimates showed that the accuracy improved for larger pixel size. However, for predicting $f_{sub} = 0.01$, the standard deviation of the confidence interval was large regardless of the size of the input image used. f_{sub} determines the proportion of the subhalo mass fraction in the lensing image, and using the lower boundary value of f_{sub} in simulating lensing images will therefore reduce the mass of subhalos in the Einstein ring such that the neural network can hardly extract subhalos features, thereby affecting the accuracy of training.

(iv) Although the SBI neural network performs well for the gravitational lens with SIS profile, the gravitational lens is more commonly described by the SIE profile. Figure 10 and Table 1 show that when the lensing images are generated with the SIE profile and specified parameters, the absolute value of the best validation loss is 0.029 ± 0.0001 , which is 20 ± 0.5 per cent greater than that for the SIS profile. The

confidence interval of the estimates shows that each estimator sample obtained by MCMC has a huge deviation, which indicates that the posterior probability inferred by SBI is unacceptable. To find an acceptable ellipticity, under the premise of ensuring the average value of the ellipticity parameters, we reduced the standard deviation of the Sérsic index, axis ratio of the mass profile, and maximum degree of ellipticity orientation manually. The decrease of these parameters tends to improve the validation loss and confidence interval of the estimates. When the standard deviation and maximum degree of orientation of ellipticity drop to a quarter, the confidence interval of the prediction for $\{0.05, -0.9\}$, $\{0.1, -1.4\}$, $\{0.1, -0.9\}$ can cover the true labels despite a still relatively large bias compared with SIS profile. In particular, under the condition of using fixed ellipticity parameters without a standard deviation, the absolute value of the validation loss is 13 ± 0.4 per cent lower than that for lensing images obtained using specified parameters of the SIE profile and 6 ± 0.5 per cent higher than that for lensing images obtained using the SIS profile, and the confidence interval for each sample of the estimator has a range comparable to that of the SIS profile. As such, when a lensing image uses the SIE profile, the contours of the halo are distorted, and differences in the degree of distortion can directly affect the accuracy of the network prediction.

A notable observation across all experiments is that constraining the slope β proves consistently harder than constraining f_{sub} . The root cause is a physical degeneracy in the lensing signal: multiple low-mass subhalos located in close proximity along the Einstein ring collectively perturb the image in a manner that closely resembles the perturbation from a single, more massive subhalo. Images generated with quite different values of β can therefore look nearly identical, and the embedding network has limited ability to break this degeneracy from a single observation. The problem is compounded at the lowest substructure fractions (e.g., $f_{\text{sub}} = 0.01$), where few subhalos contribute detectable perturbations and the information content of the image drops sharply.

Among the nuisance parameters we tested, exposure time and pixel resolution both improve posterior quality, but the effect of exposure time is markedly larger. From a practical standpoint, this is encouraging: deeper imaging of known strong lenses can be obtained simply by scheduling additional telescope time, whereas finer angular resolution would require a different instrument or optical design. Our results therefore support prioritising deep follow-up observations of substructure candidates over waiting for higher-resolution facilities.

In all experiments testing different nuisance parameters, the estimates sampled by MCMC occasionally displayed two or more confidence intervals. This observation indicates that the predicted posterior range is too large and the samples from MCMC are not sufficient to cover the full posterior probabilities. The range of the posterior is also related to the standard deviation of the estimates, with a large posterior range indicating a high uncertainty in the predicted values.

An important methodological consideration is that the validation loss may not always be a reliable indicator of posterior quality. In our experiments, we observed that the epoch yielding the lowest validation loss does not necessarily correspond to the epoch producing the most accurate posterior estimates (as measured by MAE). This discrepancy arises because the validation loss measures the classifier’s ability to distinguish parameter–image pairs on average, whereas the posterior quality depends on how well the learned ratio approximates the true likelihood-to-evidence ratio across the full parameter space. Alternative diagnostic approaches, such as monitoring the posterior MAE on a held-out test set or using rank statistics histograms to assess posterior calibration, may provide more informative convergence criteria, although they substantially increase computational cost.

Several caveats apply to the results presented above. First, every model was trained for at most 400 epochs; for the more complex datasets (e.g., SIE with full ellipticity variation), convergence may require longer training or a more carefully tuned learning-rate schedule. Second, we used the same network architecture and the same set of hyperparameters across all experimental conditions. Because optimal hyperparameters are dataset-dependent, some of our posteriors are likely under-optimised. A dedicated hyperparameter search for each condition would almost certainly improve individual results. Notwithstanding these caveats, the qualitative trends we report—posterior improvement with longer exposure time and degradation with increasing ellipticity scatter—are consistent across all experiments and are therefore unlikely to be artefacts of suboptimal training.

5.4 Sim-to-Real Gap and Model Misspecification

A further challenge concerns the applicability of our method to real observational data. All models in this work are trained exclusively on simulated images generated with specific assumptions about the lens mass profile, source light distribution, and noise properties. Real strong lensing images will inevitably contain features not present in these simulations, including correlated noise from detector readout and image processing pipelines, foreground and background contamination, complex source morphologies, and lens light contributions. Our nuisance parameter experiments demonstrate that even controlled deviations from the baseline simulation—such as adding shot noise or varying the ellipticity—can significantly degrade posterior accuracy. It is therefore likely that models trained on current simulations would require substantial refinement before producing reliable inferences from real data. This “sim-to-real” gap represents one of the most significant challenges for the practical deployment of SBI methods in observational cosmology.

Relatedly, our experiments only consider nuisance parameters that are explicitly

included in the training data. In practice, real observations will contain sources of variability that were not anticipated during training. Previous work has shown that even small unseen nuisance parameters—corresponding to changes of only a few per cent in pixel values—can introduce severe biases in the estimated posteriors. This finding underscores the importance of making training simulations as comprehensive and realistic as possible, and of developing SBI methods that are robust to model misspecification.

5.5 Summary and Future Work

In this project, we have simulated Euclid telescope images of galaxy-galaxy gravitational lensing and applied two kinds of deep learning network to detect and characterize the substructure of strong lenses image. The mass distribution of this substructure can be parameterized by $\{f_{sub}, \beta\}$. Accurate estimation of these two parameters of substructure mass is challenging. The CNN-only method has a good performance to detect macro-model parameters like Einstein radius and the total mass of lensing image (e.g. Pearson et al., 2019; Varma et al., 2020) but has rarely been used to detect the mass distribution of substructures. We compared the results obtained by CNN-only method and SBI neural network method in this paper. To make the images more realistic and to test which physical parameters of the lensing images affect the accuracy of our SBI neural network, we also tested the lensing images generated by different nuisance parameters with randomness, electronic instrument noise with different exposure times separately and the case where the galaxies were SIE profiles of different ellipticity. These experiments and results are not included in other similar studies (Vegetti et al., 2014; Hermans et al., 2020; Brehmer et al., 2019). Our SBI network using SimpleNet for NE has relatively few layers and low complexity compared to others (e.g. Brehmer et al., 2019), and also achieved acceptable accuracy. Nevertheless, when simulated lensing image adds the PSF and shot noise as nuisance parameters or using SIE profile instead of SIS profile, in experiments with specified parameters, the validation loss of the network-predicted subhalos mass parameters is difficult to converge in an acceptable range and the accuracy is significantly impacted.

Our work can be expanded further in several directions. The lensing images we use for training and testing are all single-band in this paper. In future work, we consider the use of multi-band images like RGB images as input images to the network. The telescopes of the LSST, which will be activated shortly, will produce images of galaxies in six bands (Collett, 2015), and the multi-band pictures offer additional possibilities for detecting the sensitivity of subhalo structures. Increasing the resolution of the input image has been validated in this paper to be a feasible way of improving accuracy. The real image produced by the Euclid VIS is 4000×4000 pixels, while the input image

pixels we use are far smaller. We can use higher pixel values to improve the sensitivity of the substructure features to the embedding network.

In terms of network architecture, the SimpleNet used in this paper has potential for improvement. In SBI neural networks, substructures are only detected once for all samples (Vegetti et al., 2014), and the substructure features extracted by the embedding network are sometimes insufficient to estimate f_{sub} and β . It is possible to improve the receptive field of the convolutional neural network by changing the filter size and by setting different depths and widths. We consider designing the embedding network following the structure of VGGNet (Simonyan and Zisserman, 2014) and applying other modules such as Inception (Szegedy et al., 2015) and the inverted residual block (Sandler et al., 2019).

Additionally, shot noise and PSF can be used as latent variables for SBI network training rather than nuisance parameters (Brehmer et al., 2019), an approach that has the potential to improve the accuracy of network predictions. On the training side, using a larger batch size would increase the number of contrastive atoms available at each optimisation step, potentially sharpening the learned ratio estimate. Replacing the ReLU activation with smoother alternatives (e.g., ELU) may also benefit posterior smoothness. Finally, combining posteriors from several independently trained models into an ensemble could reduce the variance of the final estimates.

Beyond the neural network architecture, several directions for improving the training data warrant investigation. The simulations used in this work do not include lens galaxy light, which would need to be subtracted or jointly modelled in real applications. Incorporating lens light would also enable multi-band analysis, where colour information may provide additional constraints on substructure. The training images also assume uncorrelated Gaussian noise, whereas real detector images exhibit correlated noise from readout electronics and drizzling during image combination. Furthermore, our simulations do not account for line-of-sight halos that are not physically associated with the lens galaxy but can produce perturbations similar to those from lens-plane subhalos. Including these effects in training simulations would be essential for applying our method to real survey data. Finally, investigating the minimum number of training samples required to marginalise over all relevant nuisance parameters simultaneously remains an open and practically important question.

6 Conclusion

Constraining the abundance and mass distribution of dark matter substructure in galaxy-scale halos is essential for testing the predictions of the Λ CDM cosmological model. In this work, we have applied and evaluated a simulation-based inference approach for estimating the subhalo population parameters $\{f_{sub}, \beta\}$ from simulated single-band

strong gravitational lensing images consistent with the Euclid space telescope.

We first demonstrated that direct regression using a CNN alone (EfficientNet) is insufficient for this task, yielding an MAE of 0.085 ± 0.0004 for 60×60 pixel images and failing entirely on images with nuisance parameters. This failure reflects the fundamental difficulty of the problem: substructure perturbations affect only a small fraction of pixels in the lensed image and cannot be reliably extracted through standard regression. By contrast, our SBI approach—which combines a lightweight embedding network (SimpleNet) with a ResNet-like ratio estimation classifier—achieves an MAE of 0.005 ± 0.0001 on the same baseline images, representing an improvement by more than an order of magnitude.

We further conducted a systematic investigation of how observational nuisance parameters affect the quality of the inferred posteriors. Our experiments show that:

- Adding random lens redshifts and source positions has a modest effect on prediction accuracy, with the validation loss increasing by only 4 ± 0.3 per cent.
- PSF convolution and shot noise degrade performance significantly at short exposure times (565 s), but the posteriors recover to acceptable accuracy at longer exposure times (5850 s).
- Increasing the input image resolution from 30×30 to 120×120 pixels consistently reduces the validation loss, confirming that higher resolution provides more information about substructure.
- Lens ellipticity, modelled through the SIE profile, introduces the most substantial challenges: with full ellipticity variation ($\sigma_m = 1$), the validation loss increases by 20 ± 0.5 per cent relative to the SIS baseline, and the posterior contours show significant broadening.

Across all experiments, constraining the mass function slope β proved consistently more challenging than constraining the mass fraction f_{sub} , which we attribute to physical degeneracies between different subhalo configurations that produce similar lensing signatures. The network also failed to provide reliable estimates at the lowest substructure fraction ($f_{\text{sub}} = 0.01$), where the lensing perturbations are too weak for the embedding network to extract meaningful features.

The key contributions of this work are threefold. First, we provide a direct comparison between CNN regression and SBI for substructure parameter estimation, demonstrating the clear superiority of the likelihood-free approach. Second, we show that a lightweight embedding network outperforms more complex architectures for this task, suggesting that training stability is more important than representational capacity in SBI applications. Third, we present the first systematic study of how multiple observational nuisance parameters—individually and in combination—affect SBI-based

substructure inference, providing practical guidance for future applications to real survey data.

Several important challenges remain before this approach can be applied to real observational data. The sim-to-real gap—arising from differences between our simplified simulations and the complexity of real lensing images—is likely the most significant barrier. Future work should focus on incorporating more realistic image features (lens galaxy light, correlated detector noise, line-of-sight halos), extending the framework to multi-band images as will be available from Euclid and the Rubin Observatory, and developing methods that are robust to model misspecification. Ultimately, combining SBI with the large samples of strong lenses expected from forthcoming wide-field surveys offers a promising path toward population-level constraints on dark matter substructure.

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